## Geometric sequences

**1** By choosing any term after the first and dividing it by the previous term.

**2**

a $-8$  b $-0.9$  c $\frac{1}{4}$

**3**

a $108, 324$  b $-768, 3072$  c $\frac{27}{64}, \frac{81}{256}$  d $\frac{81}{4}, -\frac{243}{8}$

**4**

a

i $2$  ii $2$  iii $2$

b $T_5 = -128$

**5**

a $6, 24, 96, 384$

b $7, -14, 28, -56$

c $-2, -6, -18, -54$

d $1.3, -5.2, 20.8, -83.2$

e $700\,000, 728\,000, 757\,120, 787\,404.8$

**6** Neither

**7**

a

i Geometric as terms are formed by multiplying by a constant ratio.

ii $r = -9$

b

i Arithmetic as terms are generated by adding a constant.

ii $d = 4$

**8**

a $x = 20, y = -1280$

b $a = 3, b = -\frac{3}{5}, c = \frac{3}{625}$

**9** Yes, as $\frac{t_3}{t_1} = r^2$ and $\frac{t_5}{t_3} = r^2$ so there is a common ratio of $r^2$.

**10** $T_{10} = \frac{1}{729}$

**11** $T_4 = -150$

**12** $T_3 = 501\,760$

**13** $8$

**14**

a $r = \frac{2}{3}$

b $9, 6, 4$

**15** $18, 12, 8, \frac{16}{3}, \frac{32}{9}$

**16** $\frac{5}{3}, 5, 15$

**17**

a $\sqrt{2}$

b $16\sqrt{2}$

c $n = 13$

---

## Explicit rules

**18**

a

i $3, 12, 48, 192$

ii $r = 4$

b

i $-4, 12, -36, 108$

ii $r = -3$

**19**

a $r = \pm 2$

b $a = 4$

c $T_n = 4 \times 2^{n-1}$

**20**

a

| $n$ | 1 | 2 | 3 | 4 | 10 |
|---|---|---|---|---|---|
| $T_n$ | 25 | 5 | 1 | $\frac{1}{5}$ | $\frac{1}{78\,125}$ |

b $r = \frac{1}{5}$

**21**

**a**

| $n$ | 1 | 2 | 3 | 4 | 6 |
|---|---|---|---|---|---|
| $T_n$ | $-72$ | $96$ | $-128$ | $\frac{512}{3}$ | $\frac{8192}{27}$ |

**b** $r = -\frac{4}{3}$

**22**

**a**

**i** $r = 8$

**ii** $T_n = 5 \times 8^{n-1}$

**iii** $T_{10} = 671\,088\,640$

**b**

**i** $r = -3$

**ii** $T_n = 7 \times (-3)^{n-1}$

**iii** $T_{12} = -1\,240\,029$

**c**

**i** $r = 3$

**ii** $T_n = -5 \times 3^{n-1}$

**iii** $T_{11} = -295\,245$

**d**

**i** $r = \frac{8}{3}$

**ii** $T_n = -2 \times \left(\frac{8}{3}\right)^{n-1}$

**iii** $T_7 = -\frac{8192}{81}$

**23**

**a**

| $n$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $T_n$ | $5$ | $-20$ | $80$ | $-320$ | $1280$ |

**b**

| $n$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $T_n$ | $-27$ | $-36$ | $-48$ | $-64$ | $-\frac{256}{3}$ |

**24**

**a** $r = -4$

**b** $T_n = -9 \times (-4)^{n-1}$

**25**

**a** $2\,(-8)^{n-1}$

**b** $8192, -65\,536, 524\,288$

**c** $2\,147\,483\,648$

**26**

a $-0.3 \times 5^{n-1}$

b $-187.5, -937.5, -4687.5$

## Recursive rules

**27** $T_n = \frac{1}{5} T_{n-1}, T_1 = 9000$

**28**

a $r = 4$

b $T_{n+1} = 4T_n, T_1 = 6$

**29**

a $r = \pm 4$

b $T_n = 4T_{n-1}, T_1 = 5$

c $T_n = -4T_{n-1}, T_1 = 5$

**30**

a $r = 2$

b $a = 4$

c $T_{n+1} = 2T_n, T_1 = 4$

**31**

a $r = \pm \frac{1}{5}$

b $62\,500$

c $T_n = \frac{1}{5} \times T_{n-1}, T_1 = 62\,500$

d $T_n = -\frac{1}{5} \times T_{n-1}, T_1 = 62\,500$

**32**

a $-343$

b $-2401$

c $-16\,807$

## Sequences on the coordinate plane

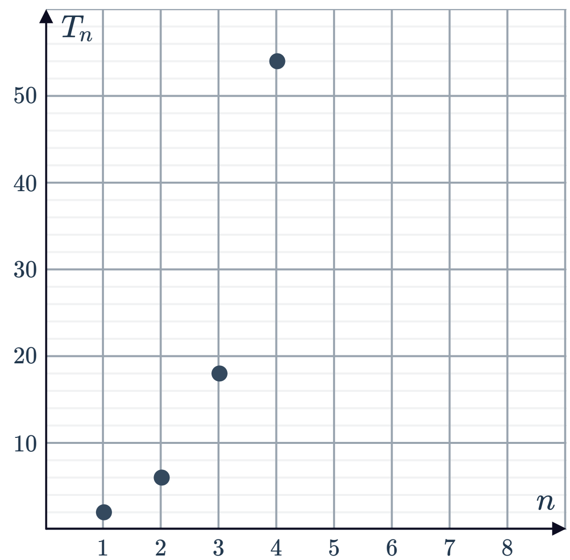
**33**

a

| $n$ | 1 | 2 | 3 | 4 | 10 |
|---|---|---|---|---|---|
| $T_n$ | 2 | 6 | 18 | 54 | 39 366 |

b $3$

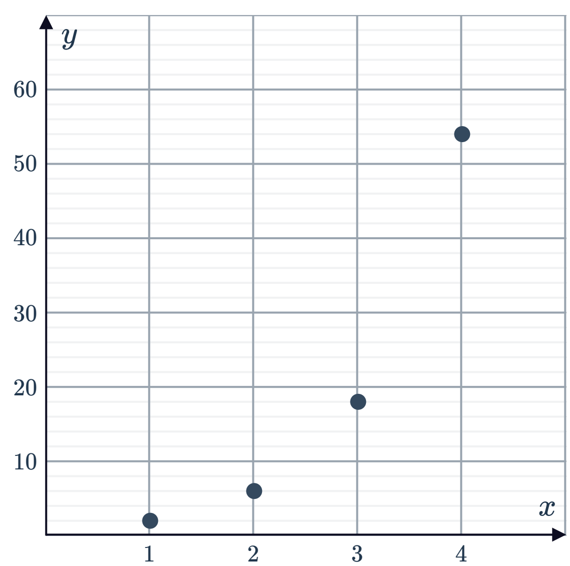
c




d Curved line

**34**

a




b Geometric

c $T_n = 3T_{n-1}, T_1 = 2$

**35**

**a**

| $n$ | 1 | 2 | 3 | 4 |
| --- | --- | --- | --- | --- |
| $T_n$ | 3 | 6 | 12 | 24 |

**b** $r = 2$

**c** $T_n = 3 \times 2^{n-1}$

**d** $T_{10} = 1536$

**36**

**a**

| $n$ | 1 | 2 | 3 | 4 |
| --- | --- | --- | --- | --- |
| $T_n$ | 2 | $-6$ | 18 | $-54$ |

**b** $r = -3$

**c** $T_4 = -54$

**d** $T_n = 2 \times (-3)^{n-1}$

**e** $T_{10} = -39\,366$

**37**

**a**

| $n$ | 1 | 2 | 3 |
| --- | --- | --- | --- |
| $T_n$ | $-2$ | $-6$ | $-18$ |

**b** $r = 3$

**c** $T_n = -2 \times 3^{n-1}$

**38**

**a** $-2$

**b** $r = 3$

**c** $T_n = -2 \times 3^{n-1}$

**39**

**a** $r = 2$

**b** $T_n = 3 \times 2^{n-1}$

**c** $T_k = -3 \times 2^{k-1}$

**40** $T_1 = 5, T_n = 3T_{n-1}$

**41**

**a** $1, 2, 4, 8, 16$

**b** Geometric, because each term increases from the last by a constant factor.

**c** $T_{n+1} = 2T_n, T_1 = 1$

## Applications

**42**

a $8.48 \text{ cm}$

b $10.10 \text{ cm}$

c $H_n = 1.06H_{n-1}, H_0 = 8$

**43**

a $\$77\,400$

b $\$57\,245.04$

c $V_n = 0.86V_{n-1}, V_0 = 90\,000$

**44**

a $\$3.74$

b $\$3.94$

c $V_n = 1.026V_{n-1}, V_0 = 3.65$

**45**

a $\$65\,536$

b $\$268\,435\,456$

c $\$8191$

d $\$536\,870\,911$

**46**

a 759 nurses

b 3980 nurses

**47**

a $12\%$

b $19\,994$

**48**

a $1:1$

b $20\,736 \text{ in}^2$

c $n = 2$

**49**

a $6832 \text{ yd}^3$

b $7168 \text{ yd}^3$

**50**

a 1216 bacteria

b 9 bacteria

**51**

a $1\,140\,152.14 \text{ m}^3$

b $1\,370\,000 \text{ m}^3$

**52**

a $1.2$

b $51.84 \text{ mm}$

**53** $n = 5$